Multi-level motion features drive animacy perception from biological motion

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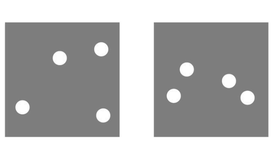
Introduction

- Animacy Perception: perceiving an entity as alive or possessing a lifelike quality.
- Biological Motion (BM), the movement produced by humans or other living creatures, can elicit rapid and spontaneous animacy perception.

Question: In the complex limb motion patterns of different species, are there any common and crucial motion features that drive animacy perception?

Methods & Results

Stimuli

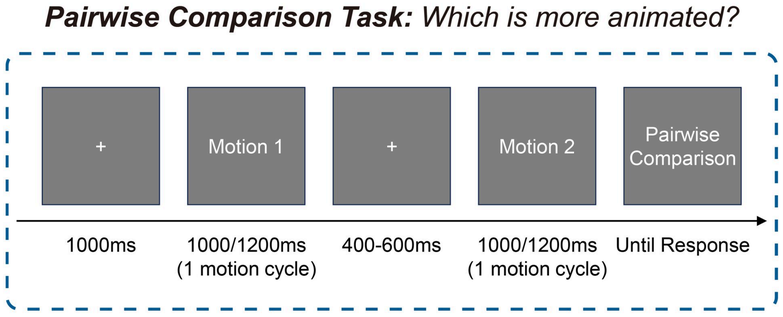


- 2 or 4 point lights: only feet, or knees and feet.
- Manipulate ***motion features*** based on natural BM.

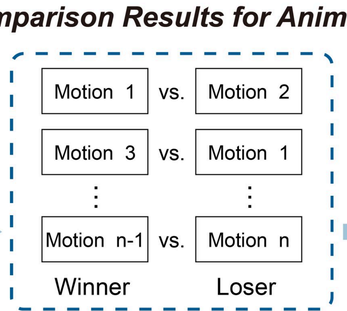
- ***Acceleration (Acc):*** low-level, kinematic
- ***Articulation (Arti):*** mid-level, rigid connection via joints
- ***Opponent Motion (Oppo):*** mid-level, anti-phase movement of contralateral limbs

Paradigm

Pairwise Comparison Task: *Which is more animated?*

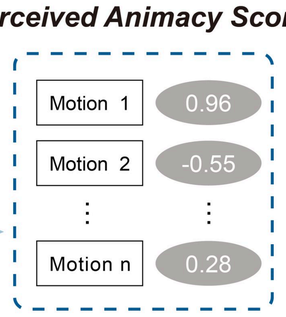


Comparison Results for Animacy



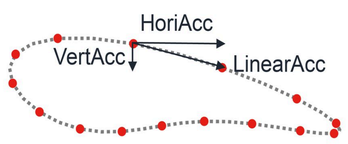
Bradley-Terry-Luce Model

Perceived Animacy Scores

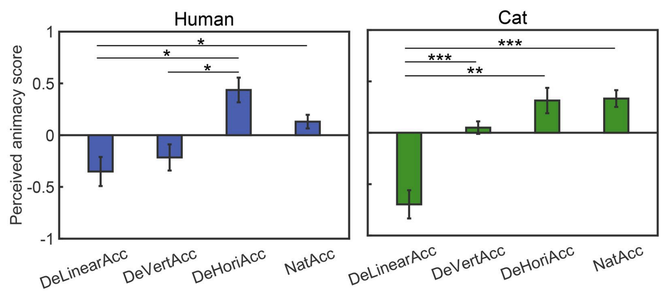


Exp.1 Foot Acceleration and its components

Foot Acc and its components:



Remove Acc components in different directions separately.



- Foot Linear Acceleration plays a key role in animacy perception.
- Horizontal Acceleration has the weakest effect in animacy perception.

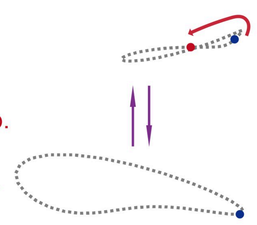
Exp.2a&2b Interaction between Articulation and Opponent Motion

Manipulation of Arti:

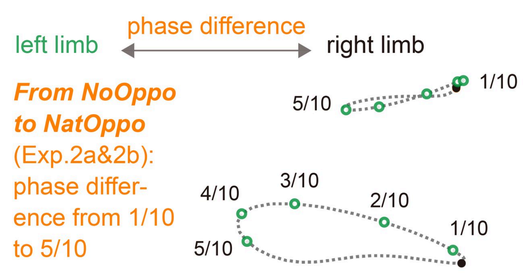
Arti (Exp.2a&2b): natural knee-foot space and phase relations.

NoArti (Exp.2b): advance knee phase by 1/4 cycle to disrupt the natural knee-foot phase relationship.

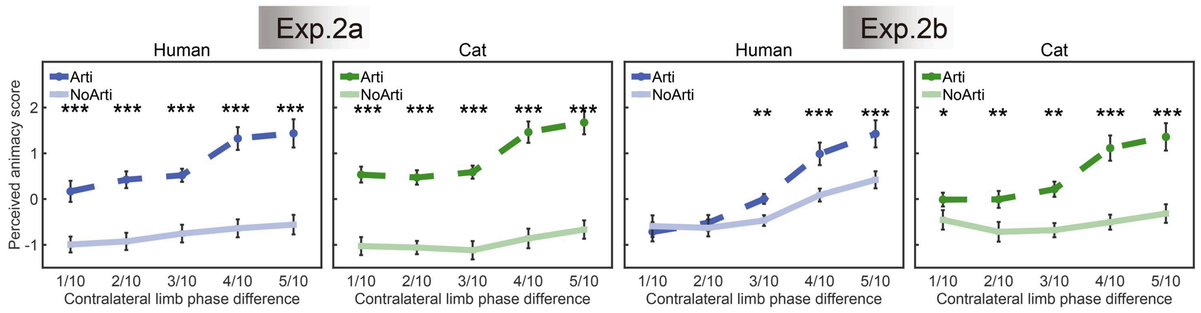
NoArti (Exp.2a): vertically swap the positions of knee and foot.



Manipulation of Oppo:



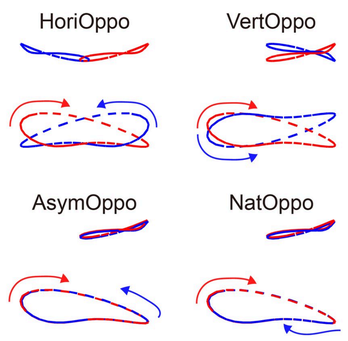
From NoOppo to NatOppo (Exp.2a&2b): phase difference from 1/10 to 5/10

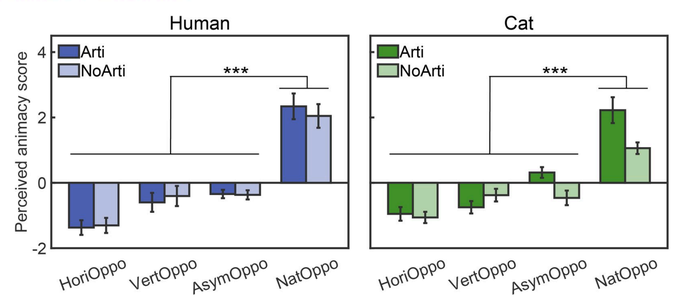


- Articulation and Opponent Motion contribute to animacy perception.
- A synergistic effect between these two features.

Exp.3a&3b Integration of Opponent Motion components

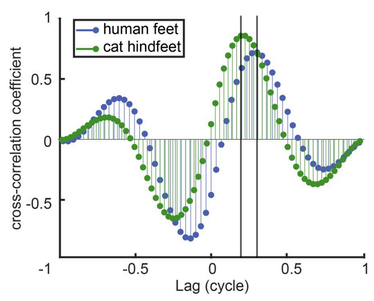
Exp.3a ***Different Oppo Components / Patterns:***





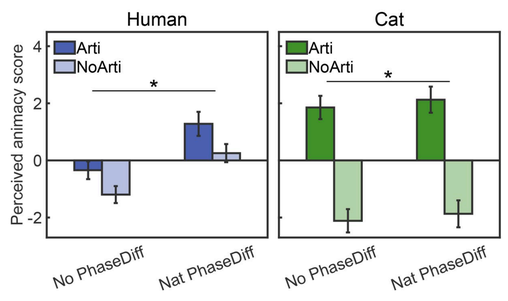
- *Horizontal and vertical Oppo components cannot drive strong percept of animacy alone or merely co-present (Asymmetrical Oppo), may require specific integration.*

Exp.3b ***Phase Difference between Hori and Vert Oppo Components:***



Natural Phase Diff:
Max cross-correlation between Hori and Vert Oppo components occurs at lag ≠ 0.

No Phase Diff:
lag = 0 by adjusting phase relation between Hori and Vert Oppo.

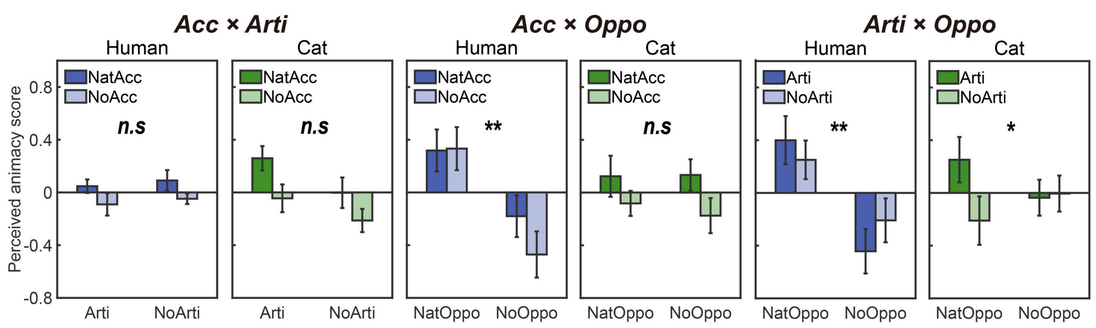


- Animacy perception from Opponent Motion requires the integration of horizontal and vertical components in a specific pattern.
- Phase difference between these two components may serve as an integration pattern significant to animacy perception.

Exp.4 Relationships between multi-level features

2 × 2 × 2 Design:

- ***NatAcc / NoAcc*** (identical to DeLinearAcc in Exp.1)
- ***Arti / NoArti*** (identical to Exp.2b&3)
- ***NatOppo / NoOppo*** (identical to 1/10 Phase Difference in Exp.2)



- Low- (Acc) and mid-level (Arti, Oppo) motion features contribute largely independently to animacy perception.

Conclusions

- Low-level (acceleration) and mid-level (articulation, opponent motion and phase difference between its components) motion features jointly drive animacy perception from cross-species biological motion.
- A synergistic effect between mid-level features and potentially distinct mechanisms for animacy perception from low- and mid-level features.
- These findings offer insights for establishing a theory of animacy perception integrating multi-level biological motion feature processing.